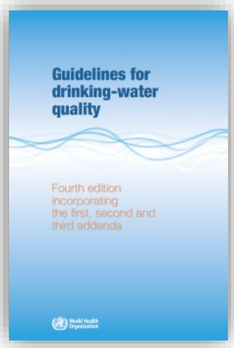
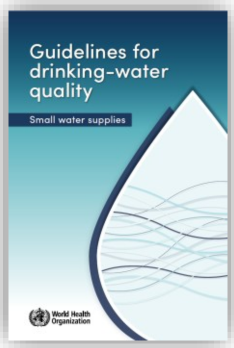
Drinking-water quality

A roadmap to supporting resources

Version: June 2026





The World Health Organization's (WHO's) Guidelines for drinking-water quality (GDWQ), and the associated Guidelines for drinking-water quality: small water supplies (GDWQ:SWS),[1] provide the international reference point for developing drinking-water quality regulations and standards.

These Guidelines present the *Framework for safe drinking-water* (the Framework) as WHO's core recommendation to ensure safe water supplies for the protection of public health. The Framework addresses the complementary functions of national regulators, water suppliers, and independent surveillance agencies (Fig. 1).

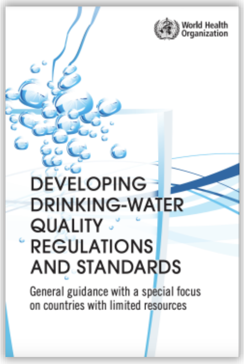
This roadmap presents a selection of resources from WHO that support implementation of the Framework, or provide the technical basis for the Guidelines recommendations. The various resources (some of which are available in multiple languages) are organized on the following pages according to the boxed headings in Fig. 1. These and other publications to support drinking-water safety can be found on WHO's Water safety and quality webpage.

Fig. 1. Structure of this roadmap document in relation to the *Framework for safe drinking-water* – WHO's key recommendation for ensuring safe drinking-water.

[1] The GDWQ:SWS offer guidance on implementing GDWQ recommendations in the context of small water supplies ranging from private household wells to community or professionally-managed piped supplies.

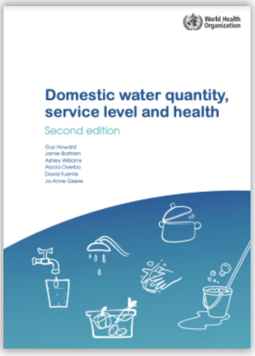
FRAMEWORK FOR SAFE DRINKING-WATER

Regulations (including health-based targets)



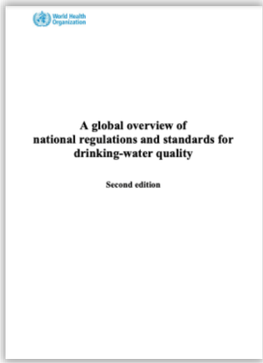
Developing drinking-water quality regulations and standards
General guidance with a special focus on countries with limited resources
WHO, 2018

Guidance on the development or revision of risk-based national or subnational drinking-water quality regulations and standards.



Domestic water quantity, service level and health
Second edition
WHO, 2020

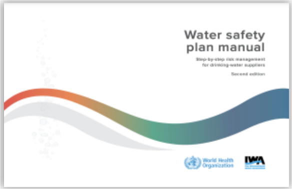
A review of the evidence on the relationships between water quantity, water accessibility and health. Also includes guidance to inform national targets on domestic water supply for health.

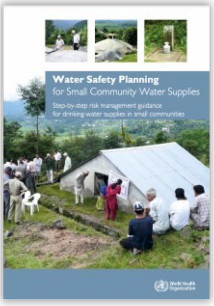


A global overview of national regulations and standards for drinking-water quality
Second edition
WHO, 2021

A summary of the water quality parameters and limits included in the drinking-water standards of 125 countries and territories. Allows regulators and other stakeholders to access and compare data when setting or revising national drinking-water standards.

Water safety plans (WSPs)

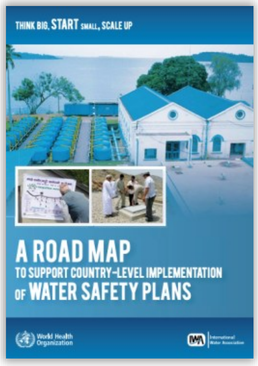




Water safety plan manual
Second edition
WHO & International Water Association (IWA), 2023

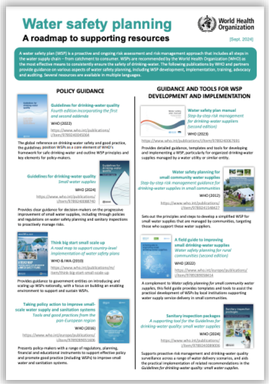
Guidance, templates and tools for developing and implementing a WSP, particularly for water supplies managed by a water utility. Supported by a training package (2026) and training videos (2021).

Tailored guidance is also provided for small community supplies (2012), which is complemented by a practical field guide (WHO/Europe, 2022).



Think big, start small, scale up
A road map to support country-level implementation of water safety plans
WHO & IWA, 2010

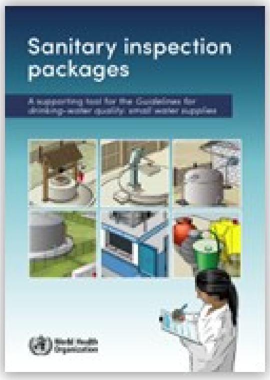
Guidance on introducing and scaling-up WSPs nationally, with a focus on building an enabling environment to support and sustain WSPs.



Water safety planning
A roadmap to supporting resources
WHO, 2024

An overview of resources available to support all aspects of water safety planning, including WSP development, implementation, training, advocacy, and auditing. Resources for key cross cutting issues are included, including using WSPs to enhance climate resilience (2017) and help ensure equity (2019).

Drinking-water quality surveillance



[Sanitary inspection packages](#)
A supporting tool for the Guidelines for drinking-water quality: small water supplies
WHO, 2024

Simple field-based checklists and associated guidance to support surveillance across a range of water delivery scenarios. Sanitary inspection can also support water safety planning and routine risk management practices by water suppliers. (*See adjacent Global Water Centre & WHO resource for a supporting training course*.)

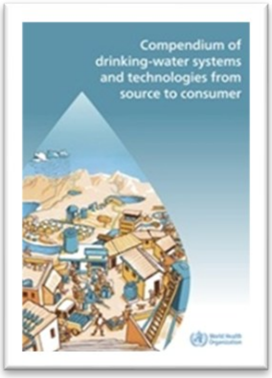
[Strengthening drinking-water surveillance using risk-based approaches](#)
WHO/Europe, 2019

Guidance on applying risk-based surveillance aligned with the WHO Framework. Outlines key principles for prioritizing surveillance activities, monitoring parameters and inspections based on local risks, WSP outcomes and public health priorities. Supported by a complementary [training package](#) (2025).

[A practical guide to auditing water safety plans](#)
WHO & IWA, 2015

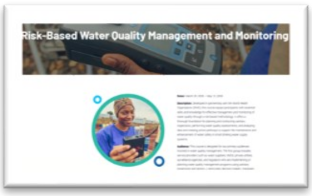
Guidance and tools on WSP auditing – a key surveillance activity – to support the continuous improvement and sustainability of WSPs, including practical examples from diverse settings. Supported by a [training package](#) (2019) and [training videos](#) (2021).

Additional supporting resources



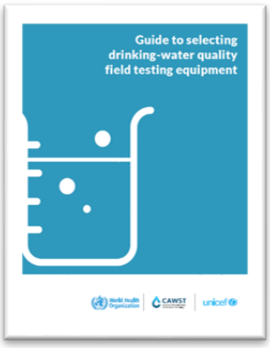
[Compendium of drinking-water systems and technologies from source to consumer](#)
WHO, 2025

Supports the selection and management of drinking-water systems and technologies, helping practitioners strengthen safe, resilient and context-appropriate services in line with WHO's Framework, including water safety planning principles.



[Risk-based water quality management and monitoring](#)
Global Water Centre & WHO, 2026

An online, facilitated course that supports implementation of WHO recommendations on risk-based principles in small systems, including planning and conducting sanitary inspections, performing water quality assessments, and analysing data for action.

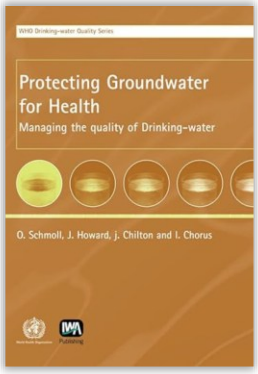


Guide to selecting drinking-water quality field testing equipment
WHO, CAWST & UNICEF (in production, expected 2026)

This guide supports the fit-for-purpose selection of drinking-water field testing equipment by helping practitioners balance considerations, including analytical performance, operational feasibility and regulatory needs to strengthen reliable water quality monitoring and public health protection.

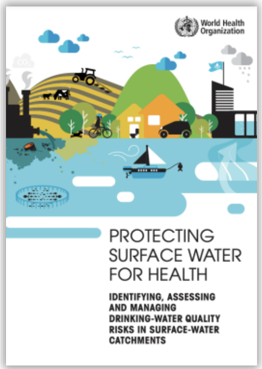
APPLICATION OF THE FRAMEWORK IN SPECIFIC CIRCUMSTANCES

Drinking-water sources



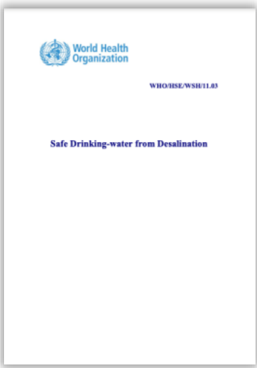
Protecting groundwater for health
Managing the quality of drinking-water
WHO, 2006

Guidance on risk assessment and risk management approaches to protect groundwater sources of drinking-water.



Protecting surface water for health
Identifying, assessing and managing drinking-water quality risks in surface-water catchments
WHO, 2016

Guidance and tools for risk assessment and risk management to protect surface water sources of drinking-water. The partner publication to Protecting groundwater for health.



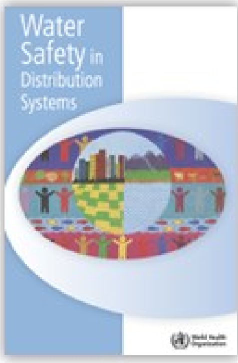
Safe drinking-water from desalination
WHO, 2011

Guidance to support risk assessment and risk management to ensure a safe drinking-water supply from desalinated sources.

Potable reuse
Guidance for producing safe drinking-water
WHO, 2017

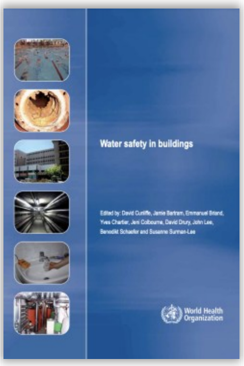
Guidance on applying appropriate management systems to produce safe drinking-water from municipal wastewater sources.

Distribution systems and buildings



Water safety in distribution systems
WHO, 2014

Guidance on enhancing risk assessment, risk management and investment planning in distribution systems.[2]



Water safety in buildings
WHO, 2011

Guidance on managing priority risks to maintain water safety within buildings (e.g. hospitals, schools, child- and aged-care facilities, hotels and apartment blocks).[2]

Health aspects of plumbing
WHO & World Plumbing Council, 2006

Guidance on the design, installation and maintenance of effective plumbing systems considering relevant microbial, chemical, physical and financial concerns.

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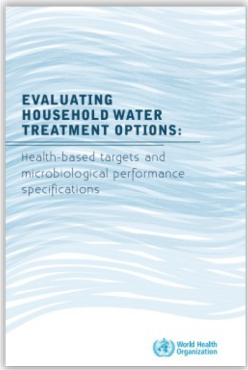
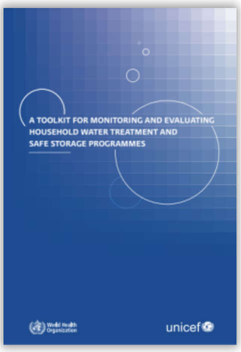
[2] The guidance presented in this publication is in line with water safety planning principles.

Households



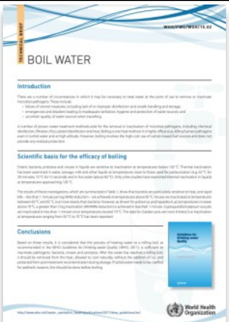
Evaluating household water treatment options
Health-based targets and microbiological performance specifications
WHO, 2011

Establishes health-based targets and microbiological performance specifications for household water treatment methods.



Toolkit for monitoring and evaluating household water treatment and safe storage
WHO & UNICEF, 2012

Guidance on planning and implementing monitoring and evaluation of household water treatment and safe storage, and on using data to promote and sustain correct practices.



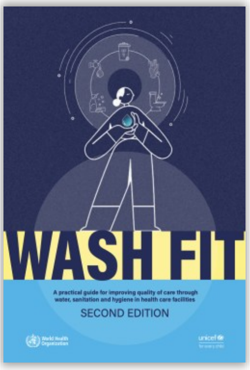
Boil water
Technical brief
WHO, 2015

A summary of the scientific basis for the efficacy of boiling as a drinking-water treatment method.

International scheme to evaluate household water treatment technologies
WHO website

A WHO scheme to independently and systematically evaluate the microbial performance of household water treatment technologies against WHO health-based criteria. Products evaluated are listed at: www.who.int/tools/international-scheme-to-evaluate-household-water-treatment-technologies/products-evaluated. The scheme website also includes resources to strengthen national capacities to evaluate and review household water treatment products, including technology specific protocols.

Health care facilities



WASH FIT
A practical guide for improving quality of care through water, sanitation and hygiene in health care facilities, second edition
WHO & UNICEF, 2022

A risk-based management tool to guide the planning and implementation of WASH improvements within health care facilities in low- and middle-income countries.

WASH in health care facilities
Knowledge portal
WHO & UNICEF website

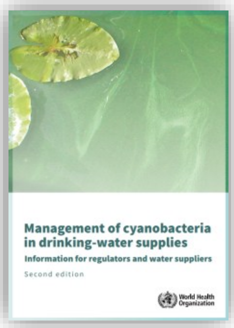
The central knowledge portal for information and resources on WASH in health care facilities, including news, publications, case studies and tools.

SUPPORTING INFORMATION UNDERPINNING THE FRAMEWORK

Chemical quality aspects

Chemical background documents
WHO website

The information source for the chemical fact sheets in the GDWQ. Reviews evidence on occurrence, potential health effects, and risk management for chemicals in drinking-water, and presents the basis for GDWQ guideline value derivation.

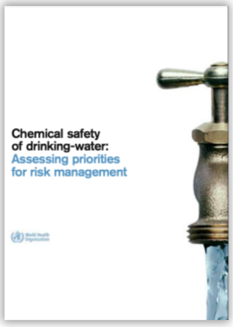


Management of cyanobacteria in drinking-water supplies
WHO, 2024

Guidance on the assessment and management of cyanobacteria and cyanotoxins in water supplies. Based on the background publication Toxic cyanobacteria in water (2021).

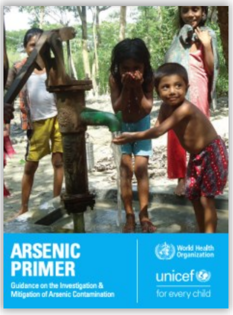
Lead in drinking-water
Health risks, monitoring and corrective actions
WHO, 2022

Guidance on the assessment and management of lead contamination in drinking-water supplies, including actions to take when elevated lead levels are detected.



Chemical safety of drinking-water
Assessing priorities for risk management
WHO, 2007

Guidance to support prioritization of chemicals for risk management and monitoring in drinking-water.



Arsenic primer
Guidance on the investigation and mitigation of arsenic contamination
WHO & UNICEF, 2018

Information on arsenic occurrence and guidance to support the development of country-specific arsenic monitoring and mitigation programmes.

Emerging issues of concern

WHO maintains a continuous review of the evidence regarding chemicals of public health concern, including pharmaceuticals (2012), microplastics (2019), chemical mixtures (2017), and a broader, ongoing initiative on PFAS.

Microbial quality aspects

[Microbiological background documents](#)
WHO website

The information source for the microbial fact sheets in the GDWQ. Reviews the evidence for water-, sanitation- and hygiene-related transmission of pathogens and provides guidance to inform risk management in water supply and sanitation systems.

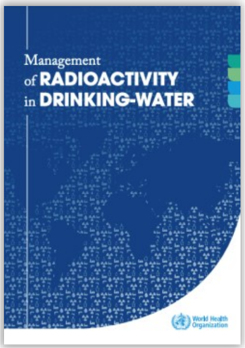
[Quantitative microbial risk assessment](#)
Application for water safety management
WHO, 2016

Guidance on the application of quantitative microbial risk assessment (QMRA) in the context of water supply, water reuse, and recreational water to support the management of microbial risks

[Water quality and health - review of turbidity](#)
Information for regulators and water suppliers
WHO, 2017

Information on the significance of turbidity in drinking-water and guidance to improve monitoring and management, including optimizing conditions for effective disinfection.

Radiological quality aspects



[Management of radioactivity in drinking-water](#)
WHO, 2018

Guidance and case studies to support the management of radionuclides in drinking-water in both emergency and non-emergency situations, building on experience from implementation of WHO guidance on radiological aspects of drinking-water quality.

Drinking-water quality and sustainable services

Safe drinking-water requires strong, sustainable water supply services, and government leadership. Governance, regulation, financing, workforce capacity, monitoring, surveillance and long-term institutional support are all important foundations for achieving and sustaining safe drinking-water, including for small water supplies. Drinking-water quality improvement efforts should therefore be considered within the broader context of strengthening sustainable and more resilient water supply systems.

For more information see State of the world's drinking water (WHO, UNICEF & World Bank, 2022) and also chapter 2 of Guidelines for drinking-water quality: small water supplies (WHO, 2024).

For more information:

Email: gdwq@who.int

Website: https://www.who.int/teams/environment-climate-change-and-health/water-sanitation-and-health/water-safety-and-quality/drinking-water-quality-guidelines